ENVIRONMENTAL IMPACT TRACKER

Calculation Data and Conversion Factor Reference — MVP Version

Purpose of this document

This document defines the data, conversion factors, fallback assumptions and formulas required to calculate a user's:

1. Water consumption
2. Electricity consumption
3. Carbon footprint

The calculation system should be deterministic. The LLM should receive the results of these calculations and evaluate them, rather than independently calculating the footprint.

The system should always use data in the following order of priority:

1. **Measured consumption**, such as litres, kilolitres, kWh or fuel litres.
2. **Consumption extracted from a bill or statement.**
3. **Consumption calculated from monetary cost using a local tariff table.**
4. **Behaviour-based estimates.**
5. **General averages or assumptions only as a final fallback.**

Each calculated result should also receive a confidence label:

Calculation method	Confidence label
Meter, bill or statement containing actual consumption	HIGH
User manually enters measured consumption	HIGH
Monetary bill converted using the correct municipal tariff	MEDIUM
Appliance or activity calculation	MEDIUM
National average or broad behavioural estimate	LOW
[Unverified MVP assumption]	VERY LOW

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1. STANDARD UNITS AND GENERAL CALCULATION RULES

Data point	Input data	Conversion factor or rule	Output / formula	Notes
Household size	Number of people	No conversion	`household_size`	Required for per-person comparisons
Monthly to annual	Monthly value	× 12	`annual = monthly × 12`	Use for bills and monthly fuel consumption
Weekly to annual	Weekly value	× 52	`annual = weekly × 52`	Use for public transport and similar activities
Daily to annual	Daily value	× 365	`annual = daily × 365`	Use for diet and daily consumption
kL to litres	Kilolitres	1 kL = 1,000 L	`litres = kL × 1000`	Standard water conversion
Tonnes CO_2e to kg CO_2e	tCO_2e	× 1,000	`kg_CO2e = tonnes_CO2e × 1000`	Keep one carbon unit internally
Percentage to fraction	Percentage	÷ 100	`fraction = percentage / 100`	Example: 35% becomes 0.35
Per-person household comparison	Household total	÷ household size	`per_person = household_total / household_size`	Do not change the actual household total

Recommended internal units

The Python calculations should standardise all results to:

- **Water:** litres per month and litres per person per day
- **Electricity:** kWh per month and kWh per year
- **Carbon:** kg CO_2e per year

2. WATER CALCULATIONS

2.1 Main water calculation table

Data point	Input data	Conversion factor or other data	Formula / output	Confidence and notes
Actual municipal water consumption	Monthly water usage in kL	1 kL = 1,000 L	`water_L_month = water_kL × 1000`	HIGH. Preferred method
Actual water consumption already in litres	Litres/ month	None	Use input directly	HIGH
Water bill amount	Rand/ month + municipality + tariff category	Use municipality-specific progressive tariff function	`water_kL = reverse_tariff(bill_amount, tariff_table)`	MEDIUM. Never use one national R-to-litre factor
National water-use fallback	Household size	218 L/person/ day	`water_L_month = 218 × household_size × 365 / 12`	LOW. Use only when no measured water data exists
Water per person per day	Household monthly litres + household size	Month normalisation	`L_person_day = water_L_month × 12 / 365 / household_size`	Use for comparisons
Shower water	Shower minutes + showers/ week	8 L/minute	`shower_L_month = minutes × showers_per_week × 8 × 52 / 12`	MEDIUM estimate
Household shower water	Per-person shower calculation	× household size	`household_shower_L = shower_L_month × household_size`	Only if shower answers describe the average individual
Rainwater share when total household water is known	Total water + rainwater percentage	Percentage ÷ 100	`rainwater_L = total_water_L × rain_fraction`	MEDIUM

Data point	Input data	Conversion factor or other data	Formula / output	Confidence and notes
Municipal share when total household use is known	Total water + rainwater share	`1 - rain_fraction`	`municipal_L = total_water_L × (1 - rain_fraction)`	Do not use this if municipal water was already measured from the municipal bill
Municipal bill plus stated rainwater share	Municipal litres + rainwater percentage	Approximation	`estimated_total_water = municipal_L / (1 - rain_fraction)`	[Approximation]. Only valid when rainwater percentage is reasonably estimated and below 100%
Garden irrigation	Yes/no or frequency category	No numerical factor in MVP	Pass as contextual variable to LLM	Do not invent litres from a yes/no answer
Swimming pool	Yes/no	No numerical factor in MVP	Pass as contextual variable to LLM	Pool water loss varies substantially with pool area, cover use, evaporation and climate

South African government sources currently use **218 litres per person per day** as the national per-capita benchmark. Eskom's residential calculator uses **8 L/minute** as its shower-flow assumption and explicitly describes its appliance calculations as approximate benchmarks rather than bill predictions.

Important rule for the coding LLM

Do not add estimated shower use on top of the 218 L/person/day national fallback.

The 218 L figure is already a total-use benchmark. Shower volume should be calculated separately to help the LLM understand where water is being used, not added to the national total.

For example:

```
estimated_total_water = 218 L/person/day baseline
estimated_shower_water = separately calculated category information
```

The LLM can then say:

> "A large portion of your estimated water use appears to come from long or frequent showers."

2.2 Converting a water bill into litres

A single rule such as:

```
R100 = X litres
```

should **not** be used across South Africa.

Municipal water tariffs can contain:

- different consumption blocks;
- fixed charges;
- free or subsidised allocations;
- different customer classifications;
- separate sanitation charges.

Current Cape Town tariff documents, for example, use several consumption bands rather than one universal rand-per-litre rate.

Recommended coding method

Store municipal tariff tables like this:

```
water_tariffs = {
    "municipality_name": {
        "fixed_charge": ...,
        "blocks": [
            {"max_kl": 6, "rate_per_kl": ...},
            {"max_kl": 10.5, "rate_per_kl": ...},
            {"max_kl": 35, "rate_per_kl": ...},
            {"max_kl": None, "rate_per_kl": ...}
        ]
    }
}
```

Then calculate the approximate consumption by reversing the progressive tariff.

For the initial prototype, it is acceptable to support only a small number of municipalities and use the behavioural fallback when the municipality is unsupported.

2.3 Optional carbon emissions associated with water

This calculation should be optional because a reliable South African national water supply-and-treatment carbon factor is not yet being used in this MVP.

Data point	Input	Factor	Formula	Notes
Water supply carbon	Municipal water in m^3	0.1913 kg CO_2e/m^3	`m3 × 0.1913`	[UK proxy]
Water treatment carbon	Wastewater volume in m^3	0.17088 kg CO_2e/m^3	`m3 × 0.17088`	[UK proxy]
Simple combined proxy	Municipal water in m^3	0.36218 kg CO_2e/m^3	`m3 × 0.36218`	[Simplified UK proxy]

These factors come from the UK Government's 2026 GHG conversion-factor dataset and should be labelled as geographic proxies, not South African measurements.

For the first version of the app, I recommend displaying water use primarily in **litres**, not adding the water carbon proxy to the user's headline carbon footprint.

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3. ELECTRICITY CALCULATIONS

3.1 Main electricity calculation table

Data point	Input data	Conversion factor or other data	Formula / output	Confidence and notes
Actual monthly electricity	kWh/month	None	Use directly	HIGH
Prepaid electricity statement	kWh purchased	None	Use actual kWh	HIGH
Electricity bill containing consumption	Extracted kWh	None	Use actual kWh	HIGH
Bill amount only	Rand + municipality/ utility + tariff	Local tariff lookup	Reverse tariff calculation	MEDIUM
Grid electricity carbon	Grid kWh	0.906 kg CO_2e/kWh	`grid_kWh × 0.906`	South African grid factor

Data point	Input data	Conversion factor or other data	Formula / output	Confidence and notes
Annual electricity carbon	Monthly grid carbon	× 12	`monthly_CO2e × 12`	Standard annualisation
Renewable share when TOTAL consumption is entered	Total kWh + renewable %	Percentage fraction	`renewable_kWh = total_kWh × renewable_fraction`	Use only when input represents whole-house energy use
Grid share when TOTAL consumption is entered	Total kWh + renewable %	`1 - renewable_fraction`	`grid_kWh = total_kWh × (1 - renewable_fraction)`	Use for grid carbon calculation
Utility bill kWh plus solar	Grid-import kWh from bill	Do not subtract solar again	`grid_kWh = bill_kWh`	The bill already represents grid imports
Grid-charged inverter or UPS	kWh already included in bill	No extra addition	Use existing bill total	Avoid double counting
Solar-charged battery backup	Estimated solar kWh	0 operational grid kWh	Keep as separate renewable category	Lifecycle panel/battery emissions excluded from MVP

The official South African 2023 grid-emission-factor report gives a national grid greenhouse-gas factor of **0.906 tCO_2e/MWh**, numerically equivalent to **0.906 kg CO_2e/kWh**.

Important rule

When a user has solar:

Do not do this:

```
electricity bill kWh × (1 - solar percentage)
```

The utility bill usually represents electricity imported from the grid. Subtracting solar from it again can undercount grid electricity.

Instead:

```
grid_kWh = measured utility kWh
solar_share = contextual information
```

When the user enters an estimate of **total whole-house electricity consumption**, the renewable percentage can be subtracted.

3.2 Electricity bill conversion

Just like water, there should not be one national rule such as:

```
R100 electricity = X kWh
```

Residential electricity pricing differs by utility, tariff category, consumption band, fixed charges and subsidy status. Current Cape Town tariffs, for example, distinguish between different residential categories and consumption structures.

Recommended priority:

```
actual kWh
    ↓
bill-extracted kWh
    ↓
municipality-specific tariff conversion
    ↓
appliance estimate
```

---

4. ELECTRICITY ESTIMATION WHEN NO BILL OR METER DATA EXISTS

Your current specification asks only about:

- home size;
- electric geyser;
- air conditioner;
- electric stove;
- pool pump.

That information is **not sufficient for a reliable deterministic calculation**.

A 1 kW appliance running for one hour and a 3 kW appliance running for eight hours are obviously very different, even though both users would answer "yes" to owning the appliance.

The simplest improvement is to add a small number of usage questions.

Data point	Input data	Conversion	Formula	Notes
Appliance electricity use	Rated watts + hours used	W ÷ 1000 = kW	`kWh = watts / 1000 × hours`	Standard formula
Monthly appliance electricity	Watts + hours/day	× 30.42 days	`kWh_month = watts / 1000 × hours_day × 30.42`	Use for AC and pumps
Pool pump	Pump wattage + hours/day	Standard kWh calculation	`pump_W / 1000 × hours_day × 30.42`	Better than yes/no
Air conditioner	Rated input power + hours/day + months/year	Standard kWh calculation	Calculate active-month consumption and annualise	Better than yes/no
Electric heater	Wattage + hours/day + months/year	Standard kWh calculation	`kW × h/day × 30.42 × months`	Use only when no measured electricity is available
Stove plate	[1.5 kW default per active plate] + minutes/day	Minutes ÷ 60	`1.5 × minutes/60 × days`	[MVP assumption] if actual rating unknown
Oven	[2.5 kW default] + hours/week	× 52	`2.5 × hours_week × 52`	[MVP assumption]
Unmeasured household base load	None	[150 kWh/month]	Add once per household	[Very rough MVP assumption]; covers refrigeration, lighting and electronics

Eskom's own residential calculator follows the same general approach: it asks about house characteristics and then requests usage information such as stove time, washing frequency, pool-pump wattage and pool-pump operating hours.

Recommendation

For the prototype, **do not attempt to make the electricity estimator highly accurate**.

Use:

```
Measured electricity available?
    YES → use it.
    NO → estimate broad consumption from major appliances.
```

The output should then be labelled:

```
calculation_confidence = "LOW"
```

---

5. GENERATORS AND BACKUP POWER

5.1 Generator emissions

Generator fuel	Input data	Conversion factor	Formula	Geographic note
Petrol	Litres/month	2.35372 kg CO_2e/L	`litres × 2.35372`	[UK Government combustion factor used as proxy]
Diesel	Litres/month	2.66155 kg CO_2e/L	`litres × 2.66155`	[100% mineral diesel proxy]
LPG	Litres/month	1.55713 kg CO_2e/L	`litres × 1.55713`	[Proxy]
Natural gas	m^3/month	2.04987 kg CO_2e/m^3	`m3 × 2.04987`	[100% mineral gas proxy]

The fuel factors are taken from the full 2026 UK Government GHG conversion-factor dataset linked by the Department for Energy Security and Net Zero. They are useful generic combustion proxies, but should not be presented as South African lifecycle emission factors.

Recommended generator input priority

Ask:

```
How many litres of fuel does the generator use in a typical month?
```

This is far better than asking only for operating hours.

If the user does not know the litres:

Input	Calculation
Generator hours/month	`fuel_L = hours × fuel_burn_rate_L_per_hour`
Manufacturer fuel rate available	Use manufacturer rate

Input	Calculation
No manufacturer information	`fuel_L = hours × [1.5 L/hour]`

The **[1.5 L/hour] value is a rough fallback assumption**, not a verified universal generator consumption rate. Generator fuel consumption varies substantially with rated power and load.

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6. PERSONAL VEHICLE CALCULATIONS

6.1 Preferred method

Data point	Input data	Conversion factor / lookup	Formula	Notes
Vehicle identification	Make + model + year	Vehicle database	Find fuel type and L/100 km	Preferred
Annual driving	km/year	None	Use directly	Required
Fuel consumption	L/100 km	÷ 100	`annual_fuel_L = annual_km × L_per_100km / 100`	Standard formula
Petrol vehicle carbon	Annual litres	2.35372 kg CO_2e/L	`annual_fuel_L × 2.35372`	[Combustion proxy]
Diesel vehicle carbon	Annual litres	2.66155 kg CO_2e/L	`annual_fuel_L × 2.66155`	[Combustion proxy]
Personal share when ride-sharing	Vehicle emissions + average occupants	Divide by average occupants	`personal_CO2e = vehicle_CO2e / occupants`	[Simplified equal-allocation method]
Electric vehicle	km/year + kWh/km	SA grid factor	`km × kWh_per_km × 0.906`	Use actual vehicle efficiency where available

The underlying combustion factors are drawn from the 2026 UK Government conversion-factor dataset, while electric-vehicle grid emissions should use the South African grid factor.

6.2 Essential fallback questions

When the vehicle database cannot find the model, ask only:

1. Fuel type
2. Average fuel consumption in L/100 km

This is much easier and safer than attempting to make the LLM estimate vehicle fuel economy from the vehicle name.

7. PUBLIC TRANSPORT

For the MVP, public transport should use passenger-kilometre factors.

Formula

```
annual_passenger_km = weekly_distance_km × 52

annual_CO2e =
annual_passenger_km
× transport_emission_factor
```

Conversion table

Transport mode	User input	Factor	Formula	Note
Bus	km/ week	0.12552 kg CO_2e/ passenger-km	`km_week × 52 × 0.12552`	[UK proxy]
Average local bus	km/ week	0.10151 kg CO_2e/ passenger-km	`km_week × 52 × 0.10151`	[UK proxy]
Coach	km/ week	0.03948 kg CO_2e/ passenger-km	`km_week × 52 × 0.03948`	[UK proxy]
Train	km/ week	0.03092 kg CO_2e/ passenger-km	`km_week × 52 × 0.03092`	[UK rail proxy; SA grid and operations differ]
Light rail / tram	km/ week	0.02121 kg CO_2e/ passenger-km	`km_week × 52 × 0.02121`	[UK proxy]
Metered taxi / ride	km/ week	0.14861 kg CO_2e/ passenger-km	`km_week × 52 × 0.14861`	[UK taxi proxy]
South African minibus taxi	km/ week	[0.12552 kg CO_2e/ passenger-km]	`km_week × 52 × 0.12552`	[Temporary bus proxy]

The numerical passenger-kilometre factors above are taken from the 2026 UK Government conversion-factor dataset and therefore need to be clearly marked as proxies.

For minibus taxis, a temporary proxy is preferable to pretending that a precise national passenger-kilometre factor exists. Recent South African research models minibus-taxi emissions using vehicle-specific fuel consumption and actual operating distances, illustrating why a more detailed future model would need occupancy and operational data.

8. FLIGHTS

8.1 Required user inputs

For every flight:

- Departure airport
- Arrival airport
- Cabin class
- One-way or return
- Number of trips per year

8.2 Airport distance calculation

The application should store:

```
airport_code
latitude
longitude
country
```

Calculate great-circle distance using the Haversine formula.

```
route_distance_km =
haversine(departure_latitude,
          departure_longitude,
          arrival_latitude,
          arrival_longitude)
```

8.3 Simple flight factors for MVP

Domestic or short-haul flight proxy

Cabin	Factor
Economy	0.12576 kg CO_2e/passenger-km
Business	0.18863 kg CO_2e/passenger-km

International non-UK flight

Cabin	Factor
Economy	0.10916 kg CO_2e/passenger-km
Premium economy	0.17465 kg CO_2e/passenger-km
Business	0.31656 kg CO_2e/passenger-km

Cabin	Factor
First	0.43663 kg CO_2e/passenger-km

Formula

```
direction_multiplier =
1 for one-way
2 for return

flight_CO2e =
route_distance_km
× cabin_factor
× direction_multiplier
× trips_per_year
```

The factors are from the UK Government's 2026 flight conversion tables. The published factors already incorporate an 8% flight-distance uplift, so the code should **not add another 8%**. The Government dataset also provides factors that include the indirect effects of non-CO_2 aviation emissions.

For the MVP:

```
same-country flight → use short-haul proxy
different-country flight → use international non-UK factor
```

This is a deliberate simplification.

---

9. HOME HEATING

The original questions:

```
heating type
months used
hours per day
```

are sufficient for electric heating **only when heater wattage is known or assumed**.

For fuels such as gas, paraffin, coal and wood, hours alone are not enough. The app should ask for the approximate amount of fuel used.

Heating method	Input data	Factor or rule	Formula	Note
Electric heater	Heater watts + hours/day + months/year	SA grid = 0.906 kg CO_2e/kWh	Calculate kWh, then × 0.906	Do not add separately if already included in measured household electricity
LPG heater	Litres/year	1.55713 kg CO_2e/L	`litres × 1.55713`	[Combustion proxy]
Natural gas	m^3/year	2.04987 kg CO_2e/m^3	`m3 × 2.04987`	[Combustion proxy]
Paraffin / kerosene	Litres/year	2.54016 kg CO_2e/L	`litres × 2.54016`	[Burning-oil proxy]
Domestic coal	kg/year	2.90495 kg CO_2e/kg	`kg × 2.90495`	[Combustion proxy]
Wood	kg/year	No simple MVP factor	Store separately and give to LLM	Biomass carbon accounting is too complicated to represent honestly as simply zero

The fossil-fuel factors are drawn from the 2026 UK Government GHG conversion-factor dataset and are geographic proxies.

---

10. DIET CARBON FOOTPRINT

For the MVP, use one annual diet factor based on the selected diet category.

App diet category	Input	Daily factor	Annual formula	Note
Heavy meat eater	Category selection	7.19 kg CO_2e/day	`7.19 × 365`	UK dietary study
Average diet	Category selection	5.63 kg CO_2e/day	`5.63 × 365`	Mapped to medium meat eater
Mostly chicken	Category selection	[4.67 kg CO_2e/day]	`4.67 × 365`	[Proxy mapping to low-meat category, not specifically chicken]
Pescatarian	Category selection	3.91 kg CO_2e/day	`3.91 × 365`	Study category: fish-eater
Vegetarian	Category selection	3.81 kg CO_2e/day	`3.81 × 365`	Study category
Vegan	Category selection	2.89 kg CO_2e/day	`2.89 × 365`	Study category

These values come from a peer-reviewed study of UK diets standardised to 2,000 kcal/day. The study reported 7.19 kg CO_2e/day for high-meat diets, 5.63 for medium-meat, 4.67 for low-meat, 3.91 for fish-eaters, 3.81 for vegetarians and 2.89 for vegans. They should therefore be treated as diet-category proxies rather than South African-specific personal measurements.

---

11. FOOD WASTE

Food-waste category mapping

User option	Waste fraction used by code
None	0.00
Very little	[0.03]
Around 10%	0.10
Around 20%	0.20
More than 30%	[0.35]

The 3% and 35% values are **[MVP assumptions]**.

Simplified calculation

```
diet_CO2_before_waste = selected daily diet factor × 365

diet_CO2_with_waste =
diet_CO2_before_waste / (1 - waste_fraction)
```

This is a **[simplified model assumption]**.

Example:

```
diet footprint = 2,000 kg CO₂e/year
food waste = 20%

adjusted footprint =
2000 / 0.8
= 2,500 kg CO₂e/year
```

The difference:

```
500 kg CO₂e
```

is attributed to the additional food that must be purchased to compensate for waste.

12. SHOPPING HABITS

The current options are:

- Much less
- Less
- Average
- More
- Much more

These are too vague to produce a scientifically defensible carbon number.

For the MVP, use an intensity score:

Shopping selection	Score
Much less	[0.60]
Less	[0.80]
Average	[1.00]
More	[1.25]
Much more	[1.50]

These scores are **[MVP assumptions]**.

Recommended use

Do:

```
shopping_intensity_score = 1.25
```

and send it to the LLM:

```
{
  "shopping_habit": "more than average",
  "shopping_intensity_score": 1.25
}
```

Do **not** do this in the MVP:

```
total_carbon += shopping_score × arbitrary tonnes of CO₂
```

The LLM can use the shopping category to recommend:

- buying fewer new clothes;
- using electronics for longer;
- repairing rather than replacing;
- buying second-hand goods;
- reducing unnecessary purchases.

A later version can ask for spending by category and use environmentally extended input-output factors.

---

13. CARBON OFFSETS

Data point	Input	Conversion	Formula
Certified offsets	tCO_2e/year	× 1,000	`offset_kg = offset_tonnes × 1000`
Gross footprint	Sum of all emission categories	None	`gross_CO2e`
Net footprint	Gross footprint minus offsets	Lower limit 0	`net_CO2e = max(gross_CO2e - offset_kg, 0)`

Always store and display:

```
gross carbon footprint
carbon offsets
net carbon footprint
```

Do not hide the actual emissions categories after subtracting offsets.

---

14. FINAL CARBON AGGREGATION

Use:

```
gross_carbon_CO2e_year =

    household_grid_electricity_CO2e
  + generator_CO2e
  + personal_vehicle_CO2e
  + public_transport_CO2e
  + flight_CO2e
  + non_electric_heating_CO2e
  + diet_and_food_waste_CO2e
```

For the first MVP:

```
shopping behaviour → qualitative analysis only
water supply carbon → optional, not included by default
renewable electricity → reduces grid electricity only where appropriate
carbon offsets → shown separately and subtracted afterwards
```

Then:

```
net_carbon_CO2e_year =
max(
    gross_carbon_CO2e_year
    - certified_offset_CO2e_year,
    0
)
```

---

15. FINAL WATER OUTPUTS

The app should send the LLM:

```
{
  "total_water_litres_month": ...,
  "municipal_water_litres_month": ...,
  "rainwater_litres_month": ...,
  "water_litres_person_day": ...,
  "shower_litres_month_estimated": ...,
  "garden_irrigation": ...,
  "pool_owned": ...,
  "calculation_method": ...,
  "confidence": ...
}
```

The most important comparison variable is:

```
water_litres_person_day
```

South Africa comparison reference:

```
218 L/person/day
```

The 218 L/person/day benchmark is the current figure stated by South African government sources.

---

16. FINAL ELECTRICITY OUTPUTS

Send:

```
{
  "total_electricity_kWh_month": ...,
  "grid_electricity_kWh_month": ...,
  "renewable_electricity_kWh_month": ...,
  "renewable_share_percent": ...,
  "electricity_CO2e_year": ...,
  "backup_power_type": ...,
  "generator_CO2e_year": ...,
  "calculation_method": ...,
  "confidence": ...
}
```

---

17. FINAL CARBON OUTPUTS

Send:

```
{
  "gross_CO2e_kg_year": ...,
  "net_CO2e_kg_year": ...,

  "breakdown": {
    "electricity": ...,
    "generator": ...,
    "personal_vehicle": ...,
    "public_transport": ...,
    "flights": ...,
    "heating": ...,
    "diet_and_food_waste": ...
  },

  "context": {
    "shopping_habit": ...,
    "shopping_intensity_score": ...,
    "garden_irrigation": ...,
    "pool_owned": ...,
    "renewable_share": ...
  },

  "carbon_offsets_kg_year": ...,
  "calculation_confidence": ...
}
```

18. MINIMUM ADDITIONAL QUESTIONS I RECOMMEND ADDING

The current project specification is largely workable, but these additions would substantially improve accuracy without making the app excessively complicated.

Section	Additional question	Why it is needed
Vehicle	Fuel type, only if database lookup fails	Prevents LLM guessing
Vehicle	L/100 km, only if database lookup fails	Allows direct carbon calculation
Pool pump	Wattage	Yes/no is insufficient
Pool pump	Hours operated per day	Required for kWh
Air conditioner	Rated power or approximate kW	Required for kWh
Air conditioner	Average hours/day when used	Required for kWh
Generator	Litres of fuel/month	Much better than hours
Fuel heating	Fuel quantity purchased per month or year	Hours alone cannot calculate emissions
Electricity bill	Does the bill amount include non-electricity charges?	Prevents bad rand-to-kWh conversions
Water bill	Is water consumption in kL already printed on the bill?	Usually better than converting the rand amount

19. VARIABLES THE LLM SHOULD RECEIVE BUT THE CALCULATOR SHOULD NOT FORCE INTO NUMERICAL EMISSIONS

These variables are still useful:

```
garden irrigation
pool ownership
home size
shopping habits
renewable technology type
backup battery type
whole-home versus essential-load backup
```

```
province
municipality
calculation confidence
```

These are valuable **context variables** for recommendations.

For example, the LLM can reason:

```
The household has high water use + a swimming pool
→ investigate pool covers and evaporation losses

High electricity + electric geyser
→ prioritise geyser optimisation or heat-pump analysis

High transport carbon + low vehicle occupancy
→ suggest carpooling

High flight footprint
→ prioritise avoiding one flight before discussing small household changes
```

This is a better use of the LLM than asking it to invent exact numerical conversions for categories where the app did not collect enough data.

---

20. RECOMMENDED FINAL SYSTEM ARCHITECTURE

Stage 1 — Collect raw user inputs

Example:

```json
{
  "household_size": 4,
  "water_kL_month": 18,
  "electricity_kWh_month": 550,
  "solar_share": 20,
  "vehicle_km_year": 15000,
  "vehicle_efficiency_L100km": 7.2,
  "fuel": "petrol",
  "diet": "vegetarian"
}
```

Stage 2 — Python performs all deterministic calculations

Example:

```
{
  "water_L_person_day": 123.3,
  "electricity_CO2e_year": 5979.6,
  "vehicle_CO2e_year": 2542.0,
  "diet_CO2e_year": 1390.7,
  "gross_CO2e_year": 9912.3
}
```

The numbers above are illustrative examples of structure, not a complete footprint calculation.

Stage 3 — Send calculated results and context to the LLM

Prompt structure:

```
You are an environmental sustainability advisor.

Do not recalculate the user's footprint.

Use the deterministic results supplied below.

Your tasks are to:

1. Explain the user's results clearly.
2. Identify the largest contributors.
3. Compare the user's behaviour with the supplied benchmarks.
4. Recommend the three to five highest-impact realistic improvements.
5. Estimate potential savings only when sufficient calculation data is
supplied.
6. Distinguish between high-confidence measured results and low-confidence
estimates.
7. Do not invent missing numerical data.
8. Prioritise major changes over trivial actions.

USER RESULTS:
{calculated_results_json}
```

---

21. SUMMARY OF WHAT SHOULD ACTUALLY BE NUMERIC IN THE MVP

Definitely calculate numerically

- Actual water consumption
- Water use per person
- Shower water estimate
- Actual electricity consumption

- Grid electricity carbon
- Generator emissions
- Personal vehicle emissions
- Public transport proxy emissions
- Flight emissions
- Fuel-based heating emissions
- Diet proxy footprint
- Food-waste adjustment
- Carbon offsets
- Gross and net carbon footprint

Keep qualitative or contextual for now

- Garden irrigation when only yes/no is known
- Swimming pool when only yes/no is known
- Shopping habits
- Home size by itself
- Wood heating unless a suitable lifecycle methodology is selected
- Battery manufacturing impact
- Solar-panel manufacturing impact

This boundary keeps the first version of the program relatively simple while preventing the app from generating numbers that appear precise but are actually based on weak assumptions.